

# **Unit 4**

## **Building E-commerce System**

- E-commerce Website/Software
  - Building Catalogs: Static, Dynamic
  - Building Shopping Cart
  - Transaction Processing
- Development of E-commerce Website/Software: Databases, Application Programs, Integration with ERP Systems, Integration with Payment Gateways
- Using Open Source CMS for Development of E-commerce Applications

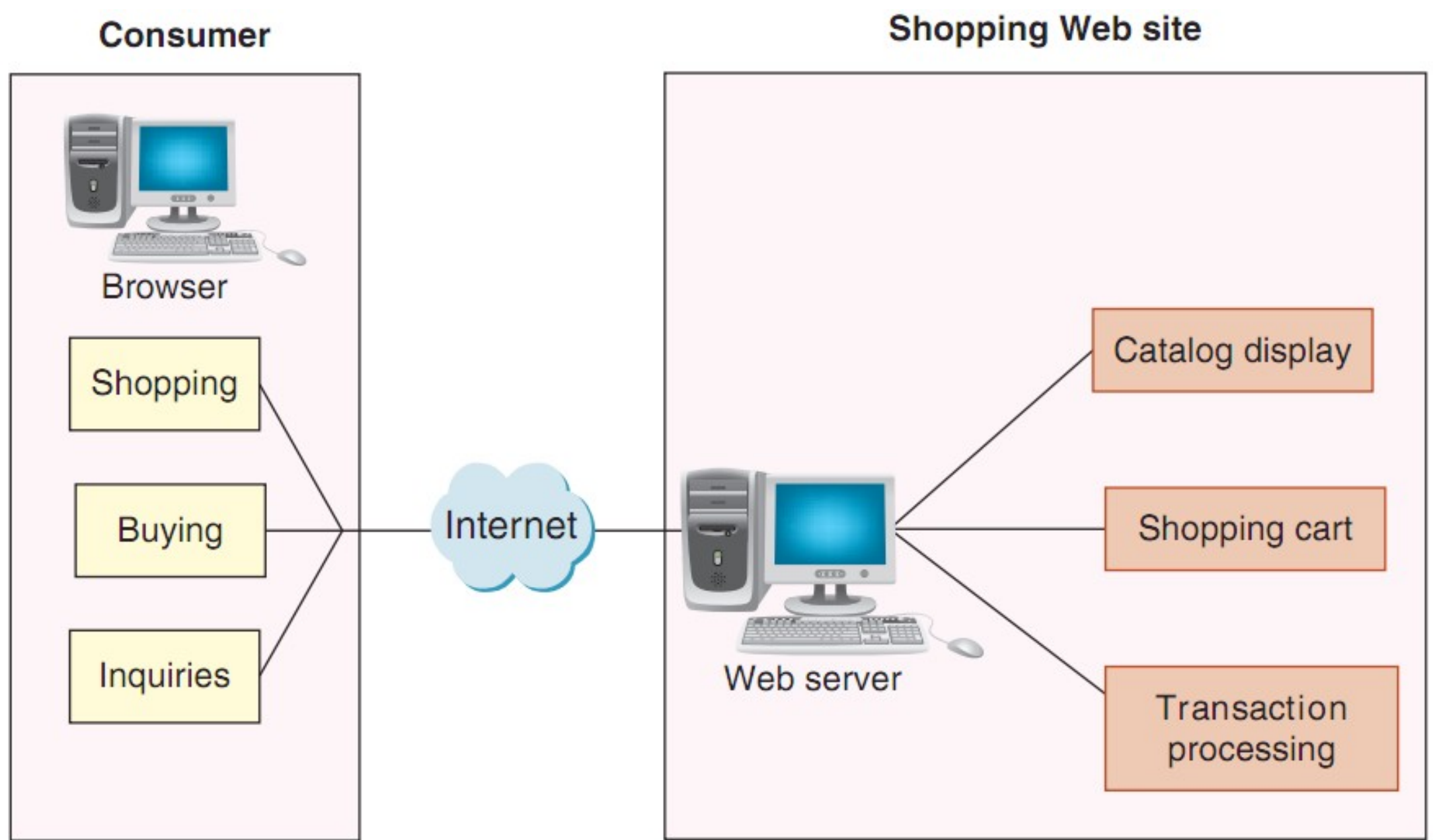
Reference: Electronic Commerce, Gary P. Schneider  
Available at : <https://szwiec.xyz/files/BIT440.pdf>

# E-commerce software

- To make E-commerce we need an infrastructure: Internet, Hardware, software, management etc...
- E-commerce software is required to make our revenue models work.
- An e-commerce software generally includes features such as catalog, shopping cart and transaction processing.
- The e-commerce software must be accompanied by a web site in order to enable users perform transactions over internet (web).
- The software and the website must be hosted on a server. There are two options for hosting:
  - self-hosting : companies can do business online using their own servers and server software.
  - third-party Web-hosting: Some companies, especially midsize and smaller companies, use this. A third-party host can provide both Web hosting and software that performs other electronic commerce functions.

# Basic Functions of Electronic Commerce Software

- Because electronic commerce sites vary in size, purpose, audience, and other factors, a range of software and hardware products are available for building electronic commerce sites.
- All electronic commerce software must provide the following elements:
  - Catalog display
  - Shopping cart capabilities
  - Transaction processing
- Larger and more complex electronic commerce sites need more complex software that adds other features and capabilities. These additional software components can include:
  - Middleware that integrates the electronic commerce system with existing company information systems that handle inventory control, order processing, and accounting
  - Enterprise application integration
  - Web services
  - Integration with enterprise resource planning (ERP) software
  - Supply chain management (SCM) software
  - Customer relationship management (CRM) software
  - Content management software
  - Knowledge management software



**FIGURE 9-2** Basic electronic commerce site architecture

# Catalog Display Software

- A catalog is an organized listing of goods or services offered for sale.
- To further organize its offerings, a retailer may break them down into departments. As in a physical store, merchandise in an online store can be grouped within logical departments to make locating an item, such as a camping stove, simpler.
- Web stores often use the same department names as their physical counterparts. In most physical stores, each product is kept in only one place. A Web store has the advantage of being able to include a single product in multiple categories. For example, running shoes could be listed as both footwear and athletic gear.

## Types of catalogs

- **Static catalog:**
  - A static catalog is a simple list, tagged with HTML codes, that can be displayed on a single Web page or a series of related Web pages.
  - This simple list approach can provide a short description and perhaps a photo of each item offered for sale. To add an item, delete an item, or change an item's listing, the company must edit the HTML-tagged text in one or more files.
  - A small online business site can have a very simple static catalog.
  - Since the catalog management can become a cumbersome process when the number of items grows beyond a few hundred, most online businesses use a dynamic catalog.

- **Dynamic catalog:**
  - A dynamic catalog stores the information about items in a database, usually on a separate computer, which is accessible to the server that is running the Web site itself.
  - A dynamic catalog can feature multiple photos of each item, detailed descriptions, and can include a search tool that allows customers to search for an item and determine its availability.
  - Dynamic catalog software is often included in the more comprehensive electronic commerce software packages
  - However, some companies write their own software to link their existing databases of product information to their Web sites.
- Besides offering a well-organized catalog, most sites with many products provide

an internal search engine, which is a software that allows customers to enter descriptive

search terms, such as “men’s shirts,” so they can quickly find the Web page containing

# Shopping Cart Software

- Helps shopper to order more than one item at a time.
- Also called a shopping bag or shopping basket, keeps track of the items customers have selected and allows customers to view the contents of their carts, add new items, or remove items.
- To order an item, the customer simply clicks a button or link near the item's description that indicates "add to cart" or other similar language.
- All of the details about the item, including its price, product number, and other identifying information, are stored automatically in the cart.
- A good shopping cart allows the customer to view the cart's contents and remove the unwanted items at any time during the session by providing a link to the cart on every page displayed during the shopping process.
- When the customer is ready to conclude the shopping session, the click of a button takes the customer to a series of checkout screens that lead to execution of the purchase transaction.

- Examples: BigCommerce, SalesCart, Volusion.
  - Some shopping cart software is sold for one-time licensing fee
  - some packages are sold on a subscription license basis (per month)
  - In some cases a transaction fee assessed as a percentage of sales.
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- HTTP messaging, which is the foundation of the Web, is a stateless system (it does not retain information from one transmission or session to another)
  - Thus shopping cart software must store information about specific shoppers and their purchases.
    - How to do it?
      - cookies



➤ Identifying number:

- Some shoppers configure their Web browser software to refuse cookies, so many sites use another method assigning each shopper a temporary identifying number. This number is added to the end of the URL that appears in the browser's address bar and persists as the shopper navigates from one Web site to another.
- this approach works only as long as the shopper continues to keep the browser software open. When a browser is closed, the identifying number is lost.
- The higher-priced shopping cart software options often include additional services along with the basic functions described above. These include :
  - dynamic pricing management
  - promotion management etc

# Transaction Processing

- Transaction processing occurs when the shopper proceeds to the virtual checkout counter by clicking a checkout button.
- Then the electronic commerce software performs any necessary calculations, such as volume discounts, sales tax, and shipping costs.
- At checkout, the customer's Web browser software and the seller's Web server software both switch into a secure state of communication.
- A basic electronic commerce software can generate reports that summarize sales and inventory shipped
- However, most midsize and larger companies use an accounting software package to record sales and inventory movements. So they integrate e-commerce software with accounting software.

- Issues:
  - Computing sales taxes and shipping costs are also important parts of online sales.
  - Sales tax rates and shipping rates can change often
  - so Web site managers must either monitor and update the rates continually or use software that updates the rates automatically.
  - Shipping companies such as FedEx and UPS offer software to shippers that integrates with electronic commerce software to ensure that the rates they have are current.
  - Other calculation complications include provisions for coupons, special promotions, and time-sensitive offers

# **Integration of e-commerce software with other softwares.**

- most large companies that have electronic commerce operations also have substantial business activity that is not related to electronic commerce.
- Thus, integrating electronic commerce activities into the company's other operations is very important.
- Some of the things to consider for integration are:
  - Database
  - Middleware
  - Enterprise applications
  - Integration with ERP systems

- **Database**

- A database is a collection of information that is stored on a computer in a highly structured way.
- The rules a business establishes about its database structure are carefully thought out and take into account how the company does business (its business rules) and how the company can reduce the likelihood that errors and inconsistencies will develop in the database.
- The most commonly used low-end database manager is Microsoft Access.
- More complex database managers that can handle larger databases and can perform more functions at higher speeds include IBM DB2, Microsoft SQL Server, and Oracle.
- Large companies use distributed database systems.
- Most companies use commercial database products, however, some use an open-source software MySQL.
- Oracle sells annual subscriptions for MySQL support and maintenance services.
- Online stores that sell many different products use databases that store product information, including size, color, type, and price details.
- Most companies that have both online and physical operations use the same database for all functions of their businesses.

- **Middleware**

- Large companies usually establish connections between their electronic commerce software (that is, their catalog display, shopping cart, and transaction processing software) and their accounting and inventory management databases or applications by using middleware.
- Middleware is software that takes information from different systems and translates to a format that other end system can understand
- e.g. middleware can take data from sales and inventory shipments from the electronic commerce software and transmits it to accounting and inventory management software in a form that these systems can read.
- Some larger companies develop their own middleware.
- however, most companies purchase middleware that is customized for their businesses by the software vendor
- Interoperability is an important goal of companies when they install middleware.
- Major middleware vendors include Broadvision, IBM Tivoli Software, and Informatica.

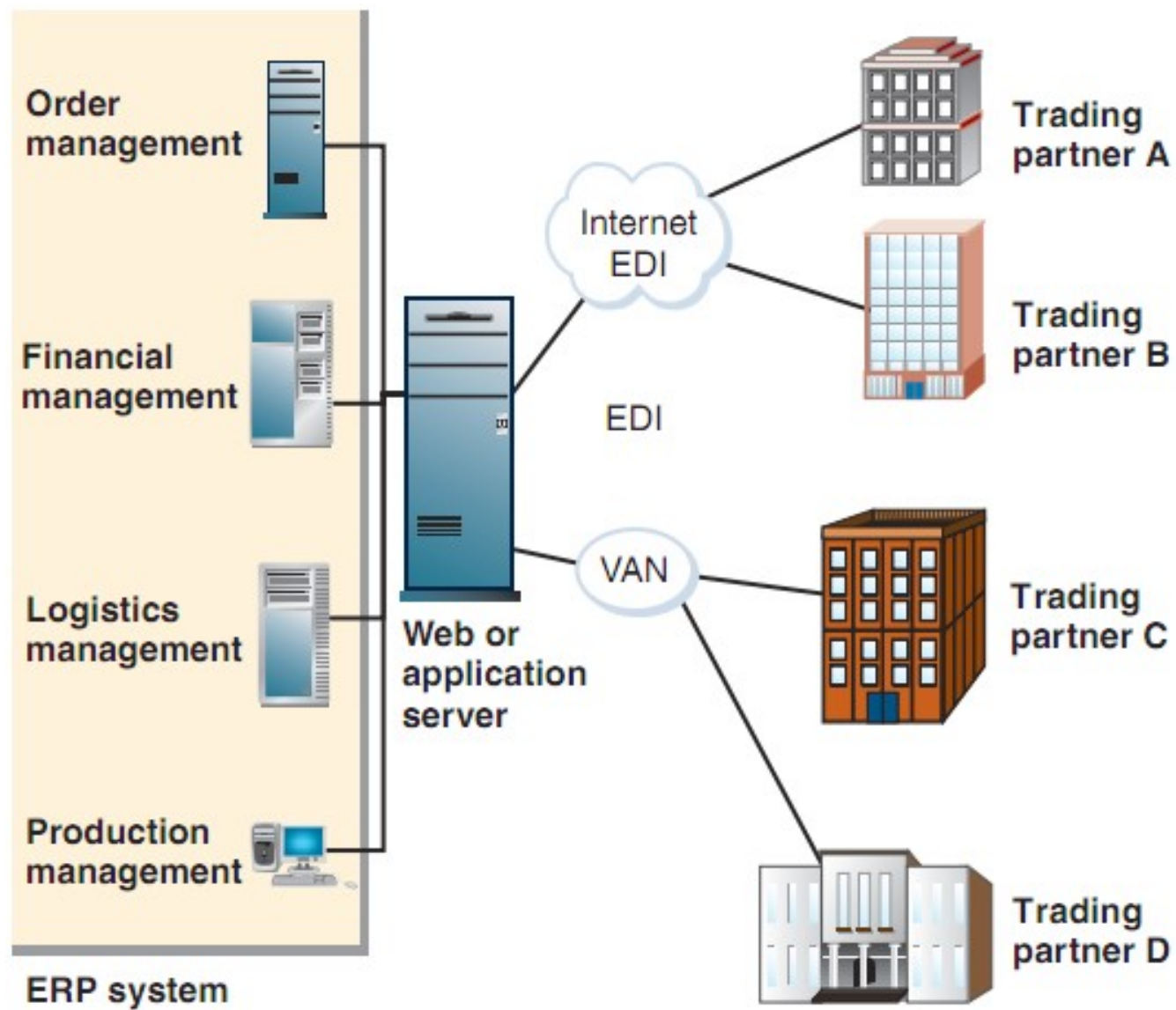
- **Enterprise Application Integration**

- A program that performs a specific function, such as creating invoices, calculating payroll, or processing payments received from customers, is called an application program, application software, or, more simply, an application.
- An application server is a computer that takes the request messages received by the Web server and runs application programs that perform some kind of action based on the contents of the request messages.
- The actions that the application server software performs are determined by the rules used in the business.
- These rules are called business logic.
- In many organizations, the business logic is distributed among many different applications that are used in different parts of the organization.
- Organizations create links among these scattered applications so that the organization's business logic can be interconnected.
- The creation and management of these links are called enterprise application integration.
- The integration is accomplished by programs that transfer information from one application to another.
- For example, a program might transfer information from order entry systems in several different divisions to a single accounts receivable and sales system that integrates all enterprise-wide sales activity.

- **Integration with ERP Systems**

- Many B2B Web sites must be able to connect to complex existing information systems such as enterprise resource planning software.
- Enterprise resource planning (ERP) software packages are business systems that integrate all facets of a business, including accounting, logistics, manufacturing, marketing, planning, project management, and treasury functions.
- E.g. one popular ERP vendor is Oracle .
- A typical installation of ERP software costs a lot (\$1 million to \$100 million)





**FIGURE 9-3** ERP system integration with EDI

- smaller online businesses cannot afford to buy a full ERP software implementation
- Some vendors provide ERP system on a subscription basis.
- Instead of installing and customizing ERP software on their own computer network, the business uses a Web browser to access the ERP software on the vendor's site. This practice of offering software use online is called software as a service (SaaS)

# **Content Management Software**

- Content management software helps companies control the large amounts of text, graphics, and media files that have become crucial to doing business.
- Most content management software includes tools that help companies manage information that was once stored in paper reports, schedules, analyses, and memos.
- In addition to the text and numbers stored in traditional databases, content management systems facilitate the storage of and access to all types of information, including images, technical drawings, geographic information, videos, audio files, and so on.
- The increased use of social media and networking as part of online business operations has made content management even more important as all kinds of Web sites now put content online.
- Content management systems help them organize, control, and move this information to and from their Web sites.

## **Essential features required for CMS**

- Before using CMS, companies should perform testing to ensure that company employees find the software's provides easy procedures for performing regular maintenance (for example, adding new categories of products and new items to existing product pages)
- The software should also facilitate typical content creation tasks, such as adding sale-item specials.
- Companies that need many different ways to access corporate information—for example, product specifications, drawings, photographs, or lab test results—often choose to manage the information and access to that information using CMS.

## **Examples**

- The leading providers of CMS include IBM and Oracle, which provide the software as components in other enterprise software packages
- several smaller companies provide stand-alone content management software.

## **Cost**

- CMS generally costs between \$50,000 and \$500,000, but it can cost three or four times that much to customize, configure, and implement.

# Open Source Content Management Systems

- Open source content management platforms are platforms with a source code that anyone can inspect, modify, or enhance.
- These tools are designed to be publicly accessible and are commonly managed and maintained by organizations.
- Some open-source CMS require some development (programming) skills to customize them according to our need.

## Examples:

- 1. Alfresco:** Alfresco is a scalable, Java-based Content Management System (CMS) that offers users the tools for imaging, web content management, team collaboration, and records management. The platform also features multi-language support.
- 2. Bitrix24:** offers a document management solution both on-prem and in the cloud. Users have the ability to move their setup to their private server whenever they want to access its source code. Additionally, the platform keeps track of the user's activity stream, enables them to chat individually and in groups, and manage their tasks and projects. Users are also able to automate their marketing strategy and design website pages.

3. Concrete5: offers an open source CMS that is licensed by MIT. The platform allows users to manipulate each individual page by clicking on edit, and then moving elements of that page around to suit their needs. Users have the ability to do this without any add-ons or extensions. Concrete5 is recommended to organizations that are looking to produce magazine experiences and general web portals. Additionally, the platform works well for users looking for drag and drop page arrangement in their CMS.
4. Drupal: Drupal's open source CMS is recommended to users who are comfortable with HTML, CSS, and PHP. The platform is strong when building up a database driven website that needs to be fast, customizable, and provide a developer-friendly CMS environment that can act as a blank canvas.
5. Joomla: It is considered one of the Best Free CMS because of its strong community and support system. Joomla offers users advanced admin features, as well as more than 7,000 extensions. The platform also provides native front-end content editing, content management flexibility, and native integration of multiple languages that makes the site's content more friendly towards non-English speakers.

Reference: <https://solutionsreview.com/content-management/the-top-free-and-open-source-content-management-platforms>